

BEROVO, North Macedonia

WMO: 135980

Lat: 41.717N

Lon: 22.850E

Elev: 836

StdP: 91.68

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-10.7	-16.7	1.0	-11.2	-13.7	1.3	-7.5	7.4	-0.3	6.1	1.2	1.1	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	31.1	17.3	29.5	16.8	28.0	16.5	19.1	26.2	18.5	25.7	17.8	25.0	2.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.6	20.5	16.2	12.8	20.0	15.5	12.2	19.4	58.1	26.2	55.7	25.9	53.5	25.0	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.9	4.8	4.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-0.9	1.4	4.8	9.2	13.3	17.1	19.6	19.6	15.4	10.0	6.5	0.8
	DBStd	7.89	5.15	4.36	3.51	3.57	2.71	2.69	2.31	2.09	3.09	3.69	3.82	4.74
	HDD10.0	1249	339	241	162	53	6	0	0	0	1	48	114	286
	HDD18.3	3250	597	474	419	273	158	54	14	11	93	258	356	542
	CDD10.0	1169	0	1	2	31	106	213	297	299	164	48	9	1
	CDD18.3	129	0	0	0	0	1	17	53	52	6	0	0	0
	CDH23.3	2161	0	0	0	6	38	352	789	793	175	8	0	0
	CDH26.7	608	0	0	0	0	3	78	241	250	37	0	0	0
(14) Wind	WSAvg	1.6	1.4	1.7	2.1	1.7	1.6	1.6	1.6	1.4	1.5	1.3	1.4	1.7
(15) Precipitation	PrecAvg	596	40	37	42	55	60	53	45	44	49	60	50	63
	PrecMax	1020	106	125	117	157	125	102	140	111	148	155	91	157
	PrecMin	296	2	4	3	11	31	17	5	4	6	3	15	0
	PrecStd	147	25	27	26	33	23	25	36	31	37	35	20	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	16.8	19.3	23.5	27.1	31.3	33.5	32.9	29.8	24.3	19.7	15.1
		MCWB	6.8	8.4	9.4	11.9	15.2	17.9	18.1	17.7	15.8	13.8	10.5	10.0
	2%	DB	10.2	13.0	16.9	20.7	24.6	28.8	31.1	31.2	27.7	21.8	17.1	11.8
		MCWB	6.3	7.2	8.1	11.1	14.4	16.7	17.4	17.1	14.9	12.4	10.3	7.9
	5%	DB	8.3	10.8	14.7	18.5	22.9	27.1	29.4	29.5	25.5	19.9	15.1	10.2
		MCWB	5.4	6.2	7.4	10.1	13.6	16.6	17.2	16.7	14.4	12.1	10.2	7.2
	10%	DB	6.4	8.7	12.3	16.6	21.2	25.5	27.8	27.8	23.3	17.4	13.3	8.2
		MCWB	4.3	5.4	6.5	9.5	13.0	16.0	16.8	16.6	14.2	11.6	9.3	5.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	9.9	10.2	13.2	17.2	19.4	20.1	20.2	17.9	15.3	12.9	10.7
		MCDB	10.8	14.2	16.2	20.0	24.1	26.3	26.9	26.4	23.7	19.2	16.4	12.9
	2%	WB	6.9	8.1	9.1	12.1	15.5	18.4	19.1	19.1	16.7	14.2	11.7	9.3
		MCDB	9.4	11.3	14.6	18.2	21.9	25.6	26.4	26.0	22.4	18.2	15.0	11.5
	5%	WB	5.7	6.7	8.3	11.2	14.6	17.6	18.5	18.4	16.0	13.3	10.7	7.7
		MCDB	7.8	10.1	12.9	16.6	20.4	24.4	25.9	25.6	20.5	17.4	14.0	9.3
	10%	WB	4.5	5.5	7.2	10.3	13.8	16.7	17.8	17.7	15.2	12.4	9.7	6.1
		MCDB	6.4	8.2	11.7	15.1	19.2	23.1	25.2	24.8	20.2	16.0	12.7	8.1
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	9.9	11.1	12.0	12.5	13.7	14.7	15.7	13.7	12.1	11.1	10.1
		MCDBR	12.3	14.1	16.3	16.0	16.3	17.1	18.0	18.9	17.9	16.9	14.6	12.9
		MCWBR	8.2	7.9	8.1	7.0	6.8	6.4	6.0	5.9	6.4	7.9	8.0	8.1
	5% WB	MCDBR	10.3	11.8	13.4	13.3	13.7	15.4	15.0	15.2	12.8	12.1	11.4	8.4
		MCWBR	7.3	7.2	7.2	6.4	6.7	6.9	6.2	6.3	6.1	6.5	6.9	5.7
(40) Clear-Sky Solar Irradiance	taub		0.305	0.332	0.373	0.426	0.419	0.409	0.406	0.418	0.397	0.362	0.333	0.309
	taud		2.415	2.344	2.262	2.116	2.205	2.267	2.275	2.239	2.278	2.369	2.416	2.417
	Ebn at Noon		842	869	869	844	859	866	864	841	833	822	799	803
	Edh at Noon		79	99	121	151	142	133	131	132	118	96	79	73
(44) All-Sky Solar Radiation	RadAvg		1.70	2.56	3.64	4.66	5.65	6.45	6.70	5.97	4.42	3.08	2.00	1.50
	RadStd		0.26	0.33	0.40	0.52	0.36	0.48	0.39	0.40	0.35	0.35	0.19	0.24

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-151	N/A	N/A

Nomenclature: See separate page